

Notional Institute of Technology, Kurukshetra

BTech(4th Sem), S-2

Subject & code: CNS (CSPC-202 & ITPC-202)

D.O.E: 06/04/2026

Time: 50 mts

Marks: 15

Attempt all the questions. All question carry equal marks.

Q1.	a) A channel has a data rate of 4 kbps and a propagation delay of 20 ms. For what range of frame sizes does stop-and-wait give an efficiency of at least 50%? b) Tabulate comparisons between <i>Stop and Wait</i> , <i>Go-back N</i> , and <i>Selective Repeat</i> flow control protocols for the parameters i.e., frame transmission, sender window size, receiver window size, efficiency and frame re-transmission.	(2, 3)
Q2.	An NIT Kurukshetra organization is granted a block of addresses with the beginning address 14.24.74.0/24. The institute needs to have 3 subblocks (i.e., <i>Administrative Block</i> , <i>Computer Centre Networking and Library</i>) of addresses to use in its three subnets: Administrative subblock of 10 addresses, Computer Centre Networking subblock of 60 addresses, and Library subblock of 120 addresses. Design the subblocks.	(5)
Q3.	a) The IPv4 address 192.168.8.28 embedded in IPv6 as ::ffff:192.168.8.28 can how we convert into hex form? b) Suppose, you are an engineer and configuring a device with a MAC address of 68:AB:74:12:34:9E on the 2001:db8:abcd:0010::/64 subnet. i. Compute the full IPv6 address using the EUI-64 format. ii. Briefly describe the process or steps used to generate the Interface Identifier from the MAC address.	(2, 2, 1)